

GOVERNMENT OF THE PUNJAB

**TECHNICAL EDUCATION & VOCATIONAL
TRAINING AUTHORITY**



**CURRICULUM FOR
AUTO ELECTRICIAN
(3 – Month Course)**

**CURRICULUM SECTION
ACADEMICS DEPARTMENT**

96-H, GULBERG-II, LAHORE

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TRAINING OBJECTIVES:

Auto mobile are increasing day by day being need of the people. In order to keep the vehicles in working position, trained persons would be required.

This curriculum is designed / developed keeping in view the job market demand to impart the skill and necessarily required theoretical knowledge to the trainees for carrying out repairing / trouble shooting and maintenance of electric circuits, battery, self starter and alternator of the vehicles to meet the need of the job market along with ethical values.: -

CURRICULUM SALIENTS

Entry Level	Middle
Total duration of course	3-Month
Total Training Hours	360-Hours
	30-Hours per week
	6-Days a Week
	5-Hrs per Day
Training Methodology	Practical 90%
	Theory 10%
Medium of instruction	Urdu / English

SKILL PROFICIENCY DETAILS:

On successful completion of this course, the trainee should be able to:-

1. Use auto electrician workshop tools and equipment efficiently
2. Maintain and charge the batteries.
3. Perform simple workshop techniques as series and parallel circuits, making tester, jumper wires, soldering, tapping, sleeving, connecting thimble etc.
4. Disassemble and assemble the self-starters and to check and its service.
5. Disassemble and assemble the AC-generators for check & service.
6. Draw the wiring diagrams of popular vehicles in the country.
7. Install components, switches, Fuses and relays.
8. Wire up the electrical circuits.
9. Locate the faults and apply the remedies for starting, charging & ignition systems.

KNOWLEDGE PROFICIENCY DETAILS:

On successful completion of this course, the trainee should be able to:-

1. Define the basic workshop rules, and safety precautions
2. Define the basic electric terms.
3. Explain the names and use of workshop tools.
4. Explain the various electrical components, their location, operation, and function.
5. Describe the construction and function of battery.
6. Describe the construction and function of self-starter.
7. Express the construction and function of Alternator.
8. Express the construction and function of voltage Regulators.
9. Explain the construction and function of Relays.
10. Explain the various electrical components
11. Define the different circuits applied in a Car wiring.
12. Understand the components of charging, ignition & starting systems.

SCHEME OF STUDIES

Auto Electrician
(3-Month) Course

Sr. No	Main Topics	Theory Hrs.	Practical Hrs.	Total Hrs.
1.	Basic knowledge	6	30	36
2.	Measuring	0	30	30
3.	Basics of electrician	0	69	69
4	Bench work	6	0	6
5	Battery	3	30	33
6	Starting system	3	30	33
7	Charging system	6	30	36
8	Ignition system	6	30	36
9	Electrical and electronic devices	6	0	6
10	Wiring circuit & wiring board	0	63	63
11.	Work Ethics	0	12	12
Total		36	324	360

DETAIL OF COURSE CONTENTS**Auto Electrician
(3-Month Course)**

Sr. No.	Detail of Topics	Theory Hrs
1	Basic Knowledge ▪ Introduction to hand tools and their use. ▪ Measuring Introductions. ▪ Safety Precautions ▪ Basic electricity.	6
2	Bench work ▪ Filing ▪ Drilling ▪ Sawing ▪ Fitting	6
3	Battery ▪ Purpose. ▪ Principle. ▪ Function. ▪ Construction	3
4	Starting System ▪ Purpose ▪ Principle ▪ Function ▪ Construction	3
5	Charging System ▪ Purpose. ▪ Principle. ▪ Function. ▪ Construction	6
6	Ignition System ▪ Purpose. ▪ Principle. ▪ Function ▪ Construction ▪ Types.	6
7	Electrical and Electronic Devices ▪ Function and Construction of ▪ Various Lights, switches, and fuses. ▪ Horn, Wiper ▪ Instrument panel lights and gauges.	6
TOTAL		36

PRACTICALS

Sr. No.	Detail of Topics	Practical Hrs
1	Basic ▪ Filling. ▪ Marking. ▪ Sawing. ▪ Drilling. ▪ Threading	30
2	Measuring ▪ Steel foot rule. ▪ Vernier caliper. ▪ Micrometer. ▪ Dial gauge. ▪ Feeler gauge	30
3	Basics of Electrician ▪ Introduction to Fasteners and materials. ▪ Demonstration with magnet Box. ▪ Insulation removal, connecting wires tapping, sleeving, connecting thimbles. ▪ Marking jumper wires and test lamp. ▪ Use of AVO meter, Use of Ampere and voltmeters in series and parallel circuit, checking voltage drop. ▪ Soldering Exercises: ▪ Fix Joint, eye Joint, and lap joint etc	69
4	Battery ▪ Cautions for battery service. ▪ Preparation of Electrolyte. ▪ Checking Specific gravity. ▪ Battery charging. ▪ Battery testing. ▪ Analysis and troubleshooting. ▪ Battery removing and replacing. ▪ Care of Batteries in stock. ▪ Making terminals of battery. ▪ Dry batteries.	30
5	Starting System ▪ Trouble shooting. ▪ Specifications. ▪ Starting system circuit on vehicle. ▪ Checking, removal and installation. ▪ Disassembly. ▪ Identification of parts. ▪ Complete checking and inspection of all components according to workshop manual.	30

	▪ Assembling of starting motor. ▪ Performance tests of motor on test bench.	
6	Charging system ▪ Trouble shooting. ▪ Specifications. ▪ Charging system checking on vehicle. ▪ Fan belt adjusting. ▪ Dismantling alternator. ▪ Parts identification. ▪ Complete inspection of parts according to workshop manual. ▪ Replacing carbon brush and bearing. ▪ Diode testing. ▪ Assembling Alternator. ▪ Double unit voltage Regulator. ▪ Identifying connections and making circuits. ▪ Installing IC regulators.	30
7	Ignition System. ▪ Trouble shooting. ▪ Spark test, inspection of high tension leads. ▪ Spark plug fouling study. ▪ Ignition coil checking. ▪ Ignition circuit checking. ▪ C.B Point replacing and gap adjusting. ▪ Setting Ignition timing.' ▪ Checking air gap and pick up coil. ▪ Dismantling, checking and fitting distributor. ▪ Fitting C.B point and condenser and starting coil on motorcycle.	30
8	Wiring Circuits on Wiring Board ▪ Reading wiring diagrams. ▪ Legal requirements. ▪ Making thimble connectors. ▪ Color codes. ▪ Head lamp circuit with relay. ▪ Aiming headlights. ▪ Parking brake circuit. ▪ Hazard warning circuit. ▪ Indicator circuit. ▪ Brake light circuit. ▪ Brake light circuit. ▪ Reverse gear light circuits. ▪ Door and roof light circuit. ▪ Horn circuit and adjustment. ▪ Fuel gauge circuit. ▪ Oil pressure light circuit.	63

	▪ Trunk and hood light. ▪ Temperature gauge circuit. ▪ Brake fluid level light circuit. ▪ Power window circuit. ▪ Wind shield and wiper motor circuit. ▪ Electric Fan circuit. ▪ Glow plug circuit. ▪ Electrical shut off valve circuit. ▪ Radio and speaker circuit. ▪ Car air conditioning and heating system wiring.	
9	Work ethics	12
TOTAL		324

LIST OF MACHINERY / EQUIPMENT / TOOLS ETC
For a class of 25 Students

Name of Trade	AUTO ELECTRICIAN
Duration of Course	3-Month

Sr. No.	Nomenclature of Equipment / Tools	Quantity
1.	Combination open end ring spanner set (6-32mm)	05 Nos.
2.	Double end off set ring spanner (6-32mm)	05 Nos.
3.	Socket set (8-32 mm)	05 Nos.
4.	Allen Key set (1.5-10mm)	05 Nos.
5.	Adjustable wrench (1')	05 Nos.
6.	Combination plier (20cm)	05 Nos.
7.	Side cutter plier (16cm)	05 Nos.
8.	Lock Plier (15cm)	05 Nos.
9.	Long Nose Plier (16cm)	05 Nos.
10.	Grip Plier (10inch)	02 Nos.
11.	Crimping Plier	02 Nos.
12.	Ball Peen Hammer (250gm, 500gm)	05 each.
13.	Plastic Hammer	05 Nos.
14.	Flat Screw driver (4' 6' 8' 12')	02 each.
15.	Philip screwdriver (3' 4' 6' 8')	02 each.
16.	Hammer screw driver set	02 Nos.

17.	Hacksaw frame (300mm)	25 Nos.
18.	File (flat)with handle (150 to 300)	25 Nos.
19.	File with handle (Round) (150 to 300)	25 Nos.
20.	Tri square (6 inches)	20 Nos.
21.	Hand drill Machine (Electric)	02 Nos.
22.	Twist drill set (1mm to 12mm)	02 Nos.
23.	Diode removing and fitting tools	01 Nos.
24.	Vernier Caliper (0-160mm)	05 Nos.
25.	Micrometer (0-25mm)	05 Nos.
26.	Dial Indicator with Stand (0.01 mm)	02 Nos.
27.	Wire gauge (standard)	02 Nos.
28.	Feeler gauge (0.5 to 1 mm with 10 blade)	05 Nos.
29.	Volt Meter (0-20 V)	05 Nos.
30.	Ampere Meter (0-25 Amp)	05 Nos.
31.	Multi-meter	02 Nos.
32.	Techno-meter & Dwell Meter (Standard 4 to 8 cylinders)	02 Nos.
33.	Hydrometer	10 Nos.
34.	Steel foot rule (30cm)	05 Nos.
35.	Ignition timing light	02 Nos.
36.	Growler Tester	02 Nos.
37.	Soldering Iron (45 W & 75 W)	10 Each.
38.	Tool kit (empty)	05 Nos.
39.	Oil Can (0.25 litter)	05 Nos.
40.	V-Blocks	04 Nos.

41.	Bench press	01 Nos.
42.	Bench voices (05" size)	05 Nos.
43.	Work benches (6'x 2.5'x 2.75')	05 Nos.
44.	Writing board	02 Nos.
45.	Battery charger	02 Nos.
46.	Battery terminal cleaner	02 Nos.
47.	Wiring Board	02 Nos.
48.	Battery capacity tester	02 Nos.
49.	Commutator hacksaw with blade different size	03 Nos.
50.	Digital Avo Meter	02 Nos.
51.	Socket screw driver (02 to 10 mm)	05 sets.
52.	Battery clamp expender (standard size)	02 Nos.
53.	Battery terminal puller	04 Nos.
54.	Condenser tester (0.02 to 0.25 uf)	02 Nos.
55.	Diode Tester	02 Nos.
56.	Tool Cabinet (18"x24"x36")	04 Nos.

REFERENCE BOOKS

Auto electrician training manual by TEVTA.

Automotive Mechanics (11th Edition by W.H Crouse)

EMPLOYABILITY OF PASS-OUTS:

The pass outs of this course can find job / employment opportunities in the following sectors / areas.

1. Authorized sales / service dealers
2. Auto Parts manufacturing industries, vendors etc.
3. Private workshops.
4. Parts manufacturing firms.
5. Repairing work shops.

MINIMUM QUALIFICATION OF TEACHER / INSTRUCTOR

2 year proficiency certificate (Auto Electrician) with 6 year relevant experience.

**STANDING OPERATING PROCEDURE FOR
EVALUATION OF SHORT COURSE STUDENTS.**

Following procedure will be followed for the evaluation of students of short courses: -

- 1 Admitted students will be registered with the Punjab Board of Technical Education Lahore within one month after the last date of admission.
- 2 The testing of the students shall be carried out as follows: -

a. Grading System (Theory & Practical).

A+	Grade from 80% and above.
A	Grade from 70% to 79%.
B	Grade from 60% to 69%
C	Grade from 50% to 59%
F	Less than 50%.
Fail	Below 40% in Theory & 50% in Practical

- Candidate has to pass both Theory & Practical

b. Attendance.

Students below 80% attendance will not be admissible to appear in examination.

c. Examining Body.

Punjab Board of Technical Education, Lahore will be the Testing and Evaluation Authority.

d. Testing.

1. Conduct. The testing shall be conducted in respective institutions under overall supervision of PBTE.

2. Methodology.

Following testing methodology will be adopted:-

(a)	Class attendance / participation	=	10%
(b)	Sessional Performance (Practical exercises/ quizzes / assignments).	=	40%
(c)	Final Exams.		
	i Theory	=	10%
	ii Practical.	=	40%
Total		=	<u>100%</u>

- 3 The institute concerned will forward the result of students to Punjab Board of Technical Education Lahore on TEV/CURR/F-1 form (Attached) within seven days of termination of course.
- 4 Punjab Board of Technical Education will process the result carrying out its scrutiny / vetting and issue certificate to successful candidates as per specimen attached.
- 5 The secretary PBTE will also coordinate for the endorsement of the said certificate by General Manager (Academic)